NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CHEMISTRY 8872/01

Paper 1 Multiple Choice **19 September 2013**

50 minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, class and tutor's name on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record you choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **14** printed pages.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 Morphine, $C_{17}H_{19}NO$, is a strong narcotic drug that doctors prescribe as a painkiller. What mass of morphine has the same number of carbon atoms as 9.0 g of CO_2?

A 0.529 g **B** 3.04 g **C** 11.2 g **D** 51.7 g

2 Carbon disulfide is a by-product found in the combustion of plastic. When 1 mole of carbon disulfide is burnt in 4 moles of oxygen, carbon dioxide and sulfur dioxide are formed.

$$CS_2(g) + 3O_2(g) \rightarrow CO_2(g) + 2SO_2(g)$$

Which of the following statements about the reaction is true?

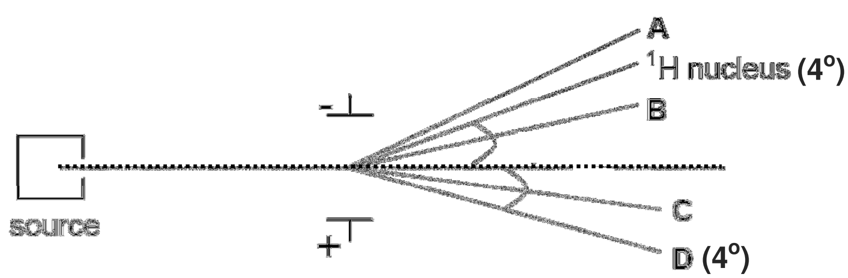
A It is a disproportionation reaction.

B Carbon in CS_2 is oxidised to CO_2.

C 6 moles of electrons are transferred during the reaction.

D When the resultant gas mixture was passed through NaOH(aq), volume decreased by 75 %.

3 When passed through an electric field, the ^{1}H nucleus is deflected as follows. Which one of the above beams represents the deflection for an ion $^{2}X^{2-}$?



4 **A** and **B** are in Group V and Group VI respectively. Which of the following comparisons between the first and second ionisation energies of **A** and **B** is correct?

	1st I.E.	**2nd I.E.**
A	A > B	A < B
B	A > B	A > B
C	A < B	A < B
D	A < B	A > B

5 In which of the following pairs of compounds is the second compound more polar than the first compound?

A	HCN	$BeCl_2$
B	NH_3	PH_3
C	SO_2	SeO_2
D	CH_3Cl	CCl_4

6 Which of the following statements about the properties associated with ionic and covalent bonds is correct?

A Only ionic compounds are soluble in water.

B All covalent compounds have low melting points.

C Both ionic and covalent bonds can occur in the same compound.

D All ionic compounds can conduct electricity under standard conditions.

7 In an experiment, 50 cm^3 of water at 30 ^oC was brought to boiling point by burning butan-1-ol in excess of oxygen. Given that the enthalpy change of combustion of butan-1-ol is $-w$ kJ mol^{-1}, calculate the percentage efficiency if m g of butan-1-ol was used in this process.

Assume that the heat capacity of water is 4.2 J K^{-1} cm^{-3} and the experiment is done under room temperature and pressure.

A $\dfrac{(50\times4.2\times70)\times74}{10mw}$ %

B $\dfrac{10mw}{10my(50\times4.2\times70)\times74}$ %

C $\dfrac{[(50+y)\times4.2\times70]\times74}{10mw}$ %

D $\dfrac{(50\times4.2\times70)\times74\times100}{mw}$ %

8 What is the value of the enthalpy change for the following process equal to?

$$Al(s) \rightarrow Al^{3+}(g) + 3e^-$$

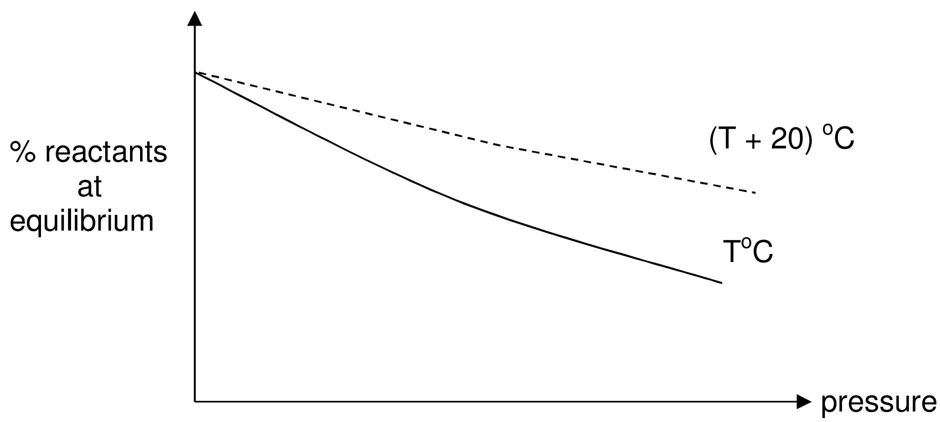
A the third ionisation energy of alumnium

B the enthalpy change of vaporisation of aluminium

C the sum of the first ionisation energy, second ionization energy and the third ionization energy of aluminium

D the sum of the enthalpy change of atomization of aluminium, the first ionisation energy, second ionization enery and the third ionization energy of aluminium

9 The graphs below show the variation of the percentage of gaseous reactants present at equilibrium, with temperature and pressure.



Which one of the following systems could the graphs represent?

A	$2Fe(s) + \frac{3}{2}O_2(g) \rightleftharpoons Fe_2O_3(s)$	$\Delta H = -822$ kJ mol^{-1}
B	$3O_2(g) + 4NH_3(g) \rightleftharpoons 2N_2(g) + 6H_2O(g)$	$\Delta H = -1248$ kJ mol^{-1}
C	$N_2O_4(g) \rightleftharpoons 2NO_2(g)$	$\Delta H = +57$ kJ mol^{-1}
D	$CO_2(g) + C(s) \rightleftharpoons 2CO(g)$	$\Delta H = +173$ kJ mol^{-1}

10 Two gases, X_2 and Y, react as follows:

$$X_2(g) + 3Y(g) \rightleftharpoons X_2Y_3(g)$$

A mixture containing 0.50 mol of X_2 and 0.50 mol of Y was heated in a 0.50 dm^3 closed container. When the reaction reached equilibrium, 0.10 mol of X_2Y_3 was produced.

What is the value of the equilibrium constant, K_c, for this reaction?

A 0.98 **B** 1.25 **C** 3.91 **D** 31.2

11 The pH of normal human blood is 7.4. Strenuous exercise can cause the condition called acidosis in which the pH falls. If the pH drops to 6.8, death may occur.

How many times greater is the hydrogen ion concentration in blood at pH 6.8 compared with that at pH 7.4?

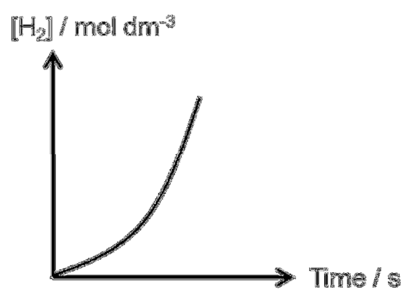
A 0.25 **B** 0.6 **C** 2.0 **D** 4.0

12 The decomposition of ammonia is a zero order reaction.

$$2NH_3\ (g) \rightarrow N_2\ (g) + 3H_2\ (g)$$

Which of these graphs correspond to the decomposition of ammonia?

A



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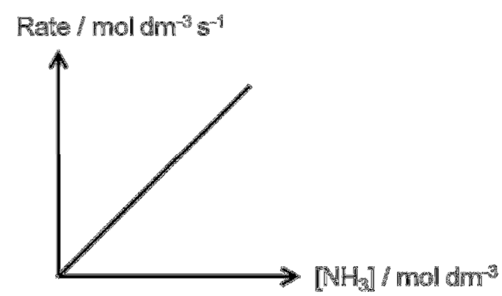
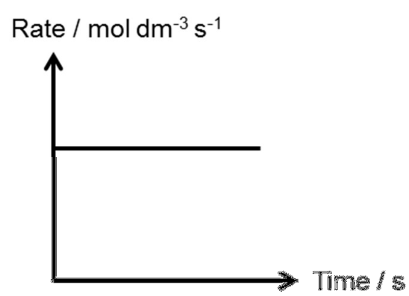
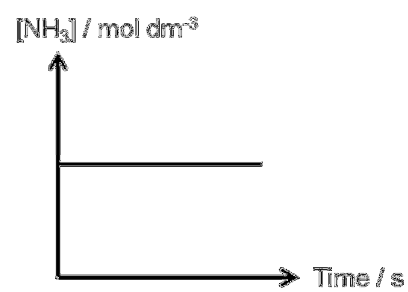


C



D



13 The decomposition of hydrogen peroxide, H_2O_2, is a first-order reaction.

It took 30 min for $[H_2O_2]$ to drop from 1.00 mol dm^{-3} to 0.25 mol dm^{-3}. At the same temperature, how long will it take for $[H_2O_2]$ to drop from 0.80 mol dm^{-3} to 0.10 mol dm^{-3}?

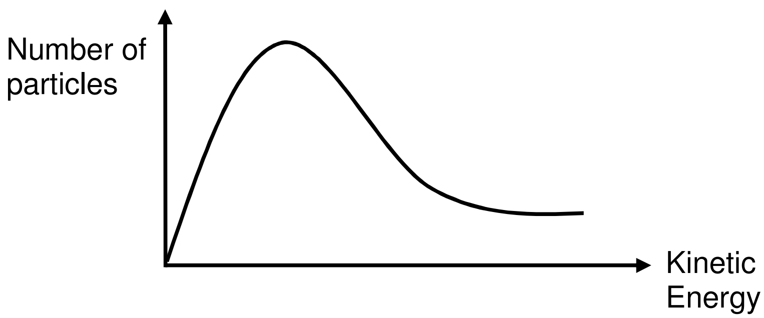
A 28 min

B 30 min

C 45 min

D 60 min

14 The diagram represents the distribution of the kinetic energy of the particles within a gas at temperature T.



Which of the following statements is correct?

A When the temperature is decreased, the area under the curve will decrease.

B When the temperature is increased, the proportion of particles with any given energy will increase.

C When the temperature is decreased, the maximum of the curve will be displaced to the right and downwards.

D When the temperature is increased, the proportion of particles with energies above any given value will increase.

15 Mixtures I, II, and III were prepared by adding 10 g of $NaCl$, $SiCl_4$ and SiO_2 respectively to 1 dm^3 of water and stirring thoroughly.

Will there be effervescence when aqueous sodium carbonate is added to these mixtures?

	I	**II**	**III**
A	no	no	yes
B	no	yes	no
C	no	yes	yes
D	yes	yes	no

16 When heated with chlorine, the hydrocarbon 3,3–dimethylpentane undergoes substitution.

How many different monochloro–substituted products can be obtained?

A 1 **B** 2 **C** 3 **D** 4

17 Given the following bond energies:

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610

Which of the following statements is true?

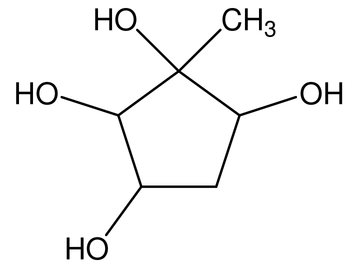
A The π bond is stronger than the σ bond.

B The C≡C bond energy is estimated to be 870 kJ mol^{-1}.

C The π bond energy cannot be determined from the data.

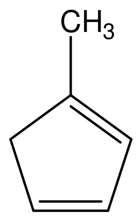
D The C≡C bond energy is the sum of the C-C bond energy and C=C bond energy.

18 A hydrocarbon on heating with cold dilute potassium manganate(VII) produces the structure below as the only organic product.

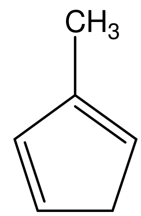


What could be a possible structure of the hydrocarbon?

A



B



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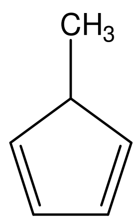

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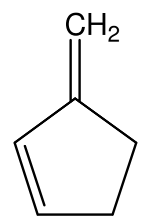


19 Ozone depletion potential (ODP) is a measure of the effectiveness of chlorofluoroalkanes in destroying stratospheric ozone.

In which sequence are compounds listed in increasing order of their ODPs?

A $CHClF_2 < CH_3CCl_2F < CCl_2FCClF_2$

B $CHClF_2 < CCl_2FCClF_2 < CH_3CCl_2F$

C $CH_3CCl_2F < CCl_2FCClF_2 < CHClF_2$

D $CCl_2FCClF_2 < CHClF_2 < CH_3CCl_2F$

20 $(CH_3)_2C(OH)COOH$, may be synthesised from $(CH_3)_2CHBr$, through a series of reactions.

Which set of reagents, used in sequential order, would be the most suitable for this synthesis?

A aqueous KOH, acidified $KMnO_4$

B HCN with NaOH catalyst, dilute H_2SO_4

C HCN with NaOH catalyst, acidified $KMnO_4$

D aqueous KOH, acidified $KMnO_4$, HCN with NaOH catalyst, dilute H_2SO_4

21 Which of the following halides will react most rapidly with aqueous sodium hydroxide?

A $(CH_3)_3CF$

B $(CH_3)_3CCl$

C $(CH_3)_3CBr$

D $(CH_3)_3CI$

22 Which of the following chemical tests can distinguish between 1-chlorobutane and 2-chlorobutane?

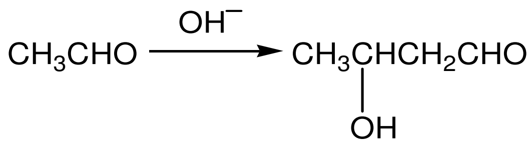
A Heating with excess NaOH(aq), followed by adding aqueous iodine.

B Heating with limited NaOH(aq), followed by adding aqueous silver nitrate.

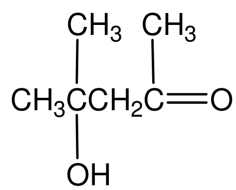
C Heating with excess NaOH(aq), followed by adding aqueous potassium dichromate (VI).

D Heating with excess NaOH(aq), followed by refluxing with acidified aqueous potassium manganate (VII).

23 In the presence of a dilute alkali, some aldehydes and ketones undergo the 'aldol reaction' where they dimerise to form a hydroxylcarbonyl compound (an aldol). For example, ethanal dimerises in this way to form 3-hydroxybutanal.



Which of the following compounds will undergo the aldol reaction to produce the aldol shown below?



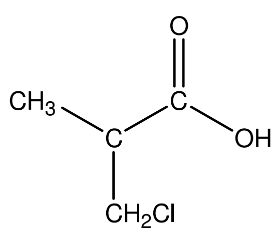
A CH_3COCH_3

B CH_3CH_2CHO

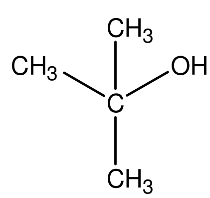
C $(CH_3)_2CHCHO$

D $CH_3CH_2COCH_3$

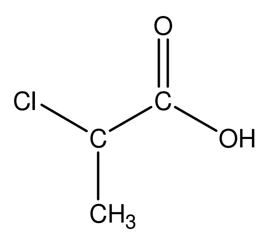
24 In which of the following sequence is the value of pKa decreasing for the following compounds?



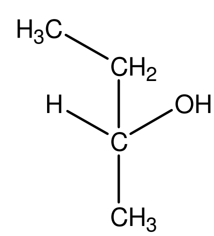
C



D



E



F

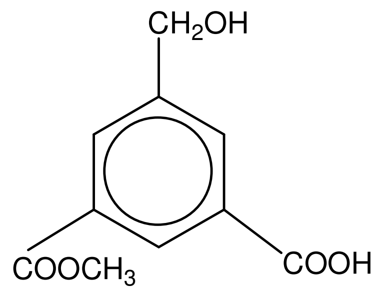
A F, D, E, C

B D, F, C, E

C C, E, D, F

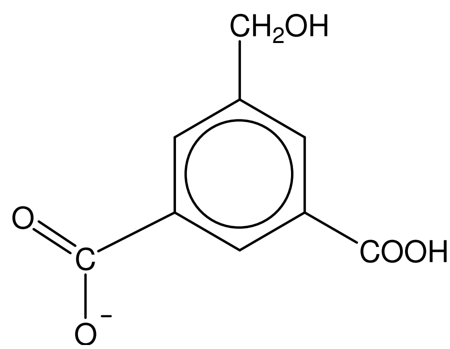
D E, C, F, D

25 Compound **G** has the following structure.

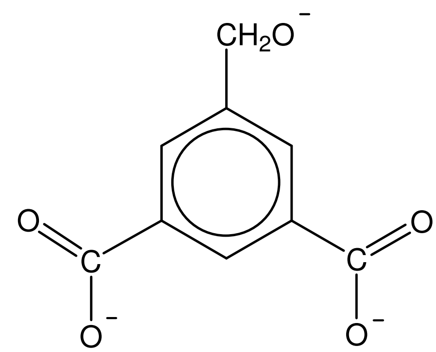


Which of the following represents the product of its reaction with NaOH (aq) at standard conditions?

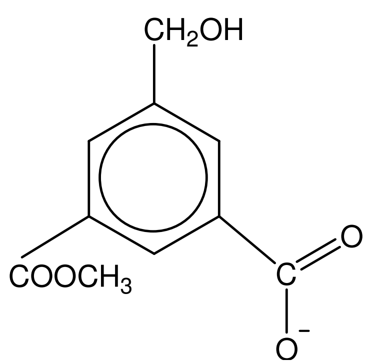
A



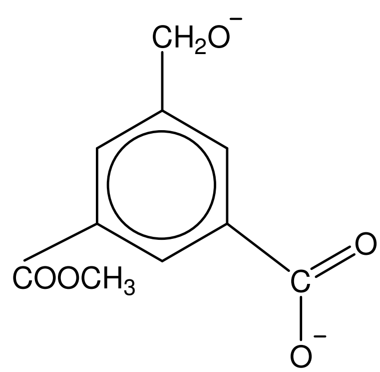
B



C



D



Section B

For each of the questions in this section one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26 Element **J** has the following successive ionisation energies in kJ mol^{-1}.

1^{st}	2nd	3rd	4^{th}	5th	6th	7^{th}
1300	3400	5300	7400	11000	13000	21000

Which of the following statements about **J** is true?

1 **J** has 2 unpaired electrons in its valence orbital.

2 **J** forms $\mathbf{J}O_2$.

3 The first IE of **J** is lower than its succeeding element.

27 Which of the following mixtures of solutions can function as a buffer?

1 50 cm^3 of 0.2 mol dm^{-3} of CH_3COOH and 25 cm^3 of 0.2 mol dm^{-3} $Ba(OH)_2$

2 40 cm^3 of 0.1 mol dm^{-3} of NH_3 and 15 cm^3 of 0.2 mol dm^{-3} HCl

3 25 cm^3 of 0.1 mol dm^{-3} of $CH_3COO^-Na^+$ and 25 cm^3 of 0.05 mol dm^{-3} HCl

28 The chloride of element **L** dissolves in aqueous NaOH and its oxide has a high melting point.

What could element **L** be?

1 aluminum

2 sodium

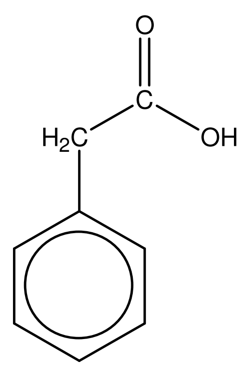
3 phosphorous

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

29 Which of the following reagents can be used to distinguish the following two compounds?



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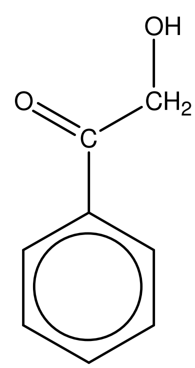


V

1 2,4-DNPH, warm

2 Na_2CO_3 (aq)

3 $NaHB_4$ (aq)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

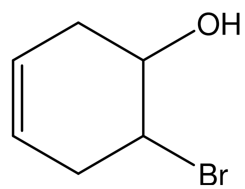
No other combination of statements is used as a correct response.

30 An organic compound **M** undergoes the following reactions:

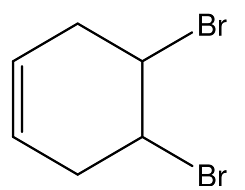
- It decolourises a solution of bromine in tetrachloromethane.
- It reacts with sodium bromide in concentrated sulfuric acid.
- It reacts with aqueous KOH to produce a compound with two alcohol functional groups.

Which compounds could be **M**?

1



2



3